

1. Buoyant motions in the atmosphere associated with local differences in heating can be vigorous. Consider a black parking lot where surface air temperature ($T = 303 \text{ K}$) is slightly warmer than that of the surrounding area ($T' = 300 \text{ K}$). What is the buoyant acceleration of the air over the parking lot?
2. When an object set to target from a distance 1000 km at 2000 km h^{-1} at the latitude of 42° , what is the displacement can be caused to the object due to Coriolis force?
3. Consider an observer fixed in space and watching the Earth rotate. What is the translational velocity of an object on the Earth at latitude 42° from the perspective of observer.
4. An air parcel with a density of 1.1 kg m^{-3} is surrounded by air with a density of 1.2 kg m^{-3} . If the volume of air parcel is 5 m^3 , calculate the buoyant force acting on the air parcel.
5. An air parcel at sea level has a temperature of 300 K . It rises to a height of 1.5 km under dry adiabatic conditions. If the dry adiabatic lapse rate is 10 K km^{-1} , what will be the temperature of the air parcel at 1.5 km ?